

Class 14: RNA-Seq Mini-Project

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Background

Our data for today comes from a HOX gene knock-out study.

Trapnell C, Hendrickson DG, Sauvageau M, Goff L et al. “Differential analysis of gene regulation at transcript resolution with RNA-seq”. Nat Biotechnol 2013 Jan;31(1):46-53. PMID: 23222703

The authors report on differential analysis of lung fibroblasts in response to loss of the developmental transcription factor HOXA1.

Section 1. Differential Expression Analysis

Data Import

We have 2 key input files: counts and metadata:

```
library(DESeq2)
```

```
Loading required package: S4Vectors
```

```
Loading required package: stats4
```

```
Loading required package: BiocGenerics
```

```
Loading required package: generics
```

```
Attaching package: 'generics'
```

```
The following objects are masked from 'package:base':
```

```
as.diffftime, as.factor, as.ordered, intersect, is.element, setdiff,  
setequal, union
```

```
Attaching package: 'BiocGenerics'
```

```
The following objects are masked from 'package:stats':
```

```
IQR, mad, sd, var, xtabs
```

```
The following objects are masked from 'package:base':
```

```
anyDuplicated, aperm, append, as.data.frame, basename, cbind,  
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,  
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,  
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,  
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,  
unsplit, which.max, which.min
```

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: GenomeInfoDb

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,

```
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars
```

Loading required package: Biobase

Welcome to Bioconductor

```
Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.
```

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

```
rowMedians
```

The following objects are masked from 'package:matrixStats':

```
anyMissing, rowMedians
```

```
metaFile <- "GSE37704_metadata.csv"
countFile <- "GSE37704_featurecounts.csv"

# Import metadata and take a peek
colData = read.csv(metaFile, row.names = 1)
head(colData)
```

```
          condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```
# Import countdata
countData = read.csv(countFile, row.names = 1)
head(countData)
```

	length	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370
ENSG00000186092	918	0	0	0	0	0
ENSG00000279928	718	0	0	0	0	0
ENSG00000279457	1982	23	28	29	29	28
ENSG00000278566	939	0	0	0	0	0
ENSG00000273547	939	0	0	0	0	0
ENSG00000187634	3214	124	123	205	207	212
	SRR493371					
ENSG00000186092	0					
ENSG00000279928	0					
ENSG00000279457	46					
ENSG00000278566	0					
ENSG00000273547	0					
ENSG00000187634	258					

Setup for DESeq Analysis

Q. Complete the code below to remove the troublesome first column from countData

```
# Note we need to remove the odd first $length col
countData <- as.matrix(countData[,-1])
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000186092	0	0	0	0	0	0
ENSG00000279928	0	0	0	0	0	0
ENSG00000279457	23	28	29	29	28	46
ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

```
rownames(colData) == colnames(countData)
```

```
[1] TRUE TRUE TRUE TRUE TRUE TRUE
```

Q. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns). Tip: What will rowSums() of countData return and how could you use it in this context?

Some genes (rows) have no count data (i.e. zero values). We should remove these before any further analysis.

```
# Filter count data where you have 0 read count across all samples.
countData <- countData[rowSums(countData) > 0, ]
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000279457	23	28	29	29	28	46
ENSG00000187634	124	123	205	207	212	258
ENSG00000188976	1637	1831	2383	1226	1326	1504
ENSG00000187961	120	153	180	236	255	357
ENSG00000187583	24	48	65	44	48	64
ENSG00000187642	4	9	16	14	16	16

Run DESeq2

```
dds = DESeqDataSetFromMatrix(countData=countData,
                              colData=colData,
                              design=~condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
dds
```

```
class: DESeqDataSet
dim: 15975 6
metadata(1): version
assays(4): counts mu H cooks
rownames(15975): ENSG00000279457 ENSG00000187634 ... ENSG00000276345
               ENSG00000271254
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(2): condition sizeFactor
```

Results

```
res = results(dds)
```

Q. Call the `summary()` function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```
summary(res)
```

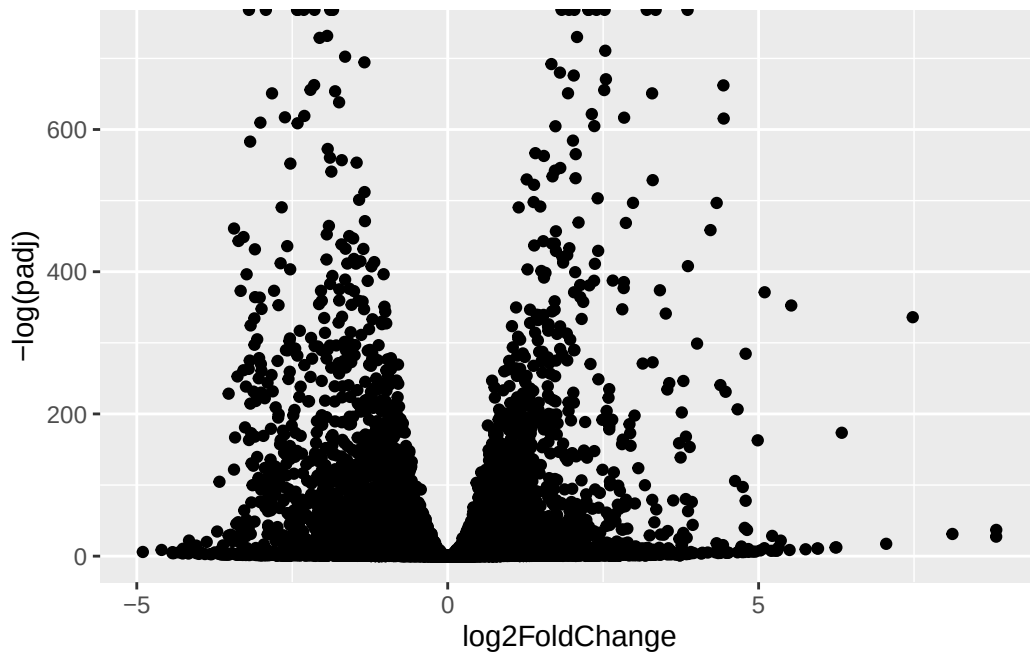
```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)      : 4349, 27%
LFC < 0 (down)    : 4396, 28%
outliers [1]      : 0, 0%
low counts [2]    : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Volcano Plot

```
library(ggplot2)

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point()
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



Q. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:

```
# Make a color vector for all genes
mycols <- rep("gray", nrow(res) )

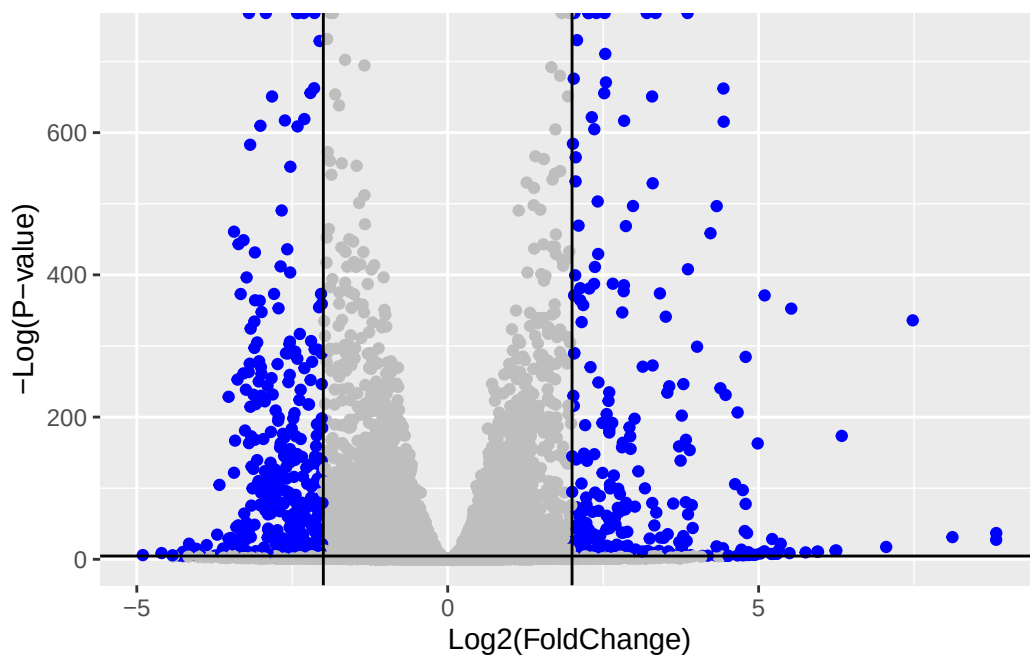
# Color blue the genes with fold change above 2
mycols[ abs(res$log2FoldChange) > 2 ] <- "blue"

# Color gray those with adjusted p-value more than 0.01
mycols[ res$padj > 0.01 ] <- "gray"
```



```
ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point(color = mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept = c(-2,2)) +
  geom_hline(yintercept = -log(0.01))
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



Adding Gene Annotation

Q. Use the `mapIDs()` function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.

```
library("AnnotationDbi")
library("org.Hs.eg.db")
```

```
columns(org.Hs.eg.db)
```

[1]	"ACCNUM"	"ALIAS"	"ENSEMBL"	"ENSEMBLPROT"	"ENSEMBLTRANS"
[6]	"ENTREZID"	"ENZYME"	"EVIDENCE"	"EVIDENCEALL"	"GENENAME"
[11]	"GENETYPE"	"GO"	"GOALL"	"IPI"	"MAP"
[16]	"OMIM"	"ONTOLOGY"	"ONTOLOGYALL"	"PATH"	"PFAM"
[21]	"PMID"	"PROSITE"	"REFSEQ"	"SYMBOL"	"UCSCKG"
[26]	"UNIPROT"				

```
res$symbol = mapIds(org.Hs.eg.db,  
  keys=row.names(res),  
  keytype="ENSEMBL",  
  column="SYMBOL",  
  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez = mapIds(org.Hs.eg.db,  
  keys=row.names(res),  
  keytype="ENSEMBL",  
  column="ENTREZID",  
  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$name = mapIds(org.Hs.eg.db,  
  keys=row.names(res),  
  keytype="ENSEMBL",  
  column="GENENAME",  
  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res, 10)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 10 rows and 9 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.913579	0.1792571	0.3248216	0.551863	5.81042e-01
ENSG00000187634	183.229650	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.188076	-0.6927205	0.0548465	-12.630158	1.43990e-36
ENSG00000187961	209.637938	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.255123	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.979750	0.5428105	0.5215598	1.040744	2.97994e-01
ENSG00000188290	108.922128	2.0570638	0.1969053	10.446970	1.51282e-25
ENSG00000187608	350.716868	0.2573837	0.1027266	2.505522	1.22271e-02
ENSG00000188157	9128.439422	0.3899088	0.0467163	8.346304	7.04321e-17
ENSG00000237330	0.158192	0.7859552	4.0804729	0.192614	8.47261e-01
	padj	symbol	entrez	name	
	<numeric>	<character>	<character>	<character>	
ENSG00000279457	6.86555e-01	NA	NA	NA	
ENSG00000187634	5.15718e-03	SAMD11	148398	sterile alpha motif ..	
ENSG00000188976	1.76549e-35	NOC2L	26155	NOC2 like nucleolar ..	
ENSG00000187961	1.13413e-07	KLHL17	339451	kelch like family me..	
ENSG00000187583	9.19031e-01	PLEKHN1	84069	pleckstrin homology ..	
ENSG00000187642	4.03379e-01	PERM1	84808	PPARGC1 and ESRR ind..	
ENSG00000188290	1.30538e-24	HES4	57801	hes family bHLH tran..	
ENSG00000187608	2.37452e-02	ISG15	9636	ISG15 ubiquitin like..	
ENSG00000188157	4.21963e-16	AGRN	375790	agrin	
ENSG00000237330	NA	RNF223	401934	ring finger protein ..	

Q. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```
res = res[order(res$pvalue),]  
write.csv(res, file="deseq_results.csv")
```

Section 2: Pathway Analysis

KEGG Pathways

```
library(pathview)  
library(gage)
```

```
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

# Focus on signaling and metabolic pathways only
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]

# Examine the first 3 pathways
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"
[49] "8824" "8833" "9" "978"
```

```
$`hsa00230 Purine metabolism`
```

```
[1] "100" "10201" "10606" "10621" "10622" "10623" "107" "10714"
[9] "108" "10846" "109" "111" "11128" "11164" "112" "113"
[17] "114" "115" "122481" "122622" "124583" "132" "158" "159"
[25] "1633" "171568" "1716" "196883" "203" "204" "205" "221823"
[33] "2272" "22978" "23649" "246721" "25885" "2618" "26289" "270"
[41] "271" "27115" "272" "2766" "2977" "2982" "2983" "2984"
[49] "2986" "2987" "29922" "3000" "30833" "30834" "318" "3251"
[57] "353" "3614" "3615" "3704" "377841" "471" "4830" "4831"
[65] "4832" "4833" "4860" "4881" "4882" "4907" "50484" "50940"
[73] "51082" "51251" "51292" "5136" "5137" "5138" "5139" "5140"
[81] "5141" "5142" "5143" "5144" "5145" "5146" "5147" "5148"
[89] "5149" "5150" "5151" "5152" "5153" "5158" "5167" "5169"
[97] "51728" "5198" "5236" "5313" "5315" "53343" "54107" "5422"
[105] "5424" "5425" "5426" "5427" "5430" "5431" "5432" "5433"
[113] "5434" "5435" "5436" "5437" "5438" "5439" "5440" "5441"
[121] "5471" "548644" "55276" "5557" "5558" "55703" "55811" "55821"
[129] "5631" "5634" "56655" "56953" "56985" "57804" "58497" "6240"
[137] "6241" "64425" "646625" "654364" "661" "7498" "8382" "84172"
```

```
[145] "84265" "84284" "84618" "8622" "8654" "87178" "8833" "9060"
[153] "9061" "93034" "953" "9533" "954" "955" "956" "957"
[161] "9583" "9615"
```

Gage Function requires the names of the values of Entrez gene IDs

```
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

```
      1266      54855      1465      2034      2150      6659
-2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288
```

```
# Get the results
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

```
attributes(keggres)
```

```
$names
[1] "greater" "less" "stats"
```

```
# Look at the first few down (less) pathways
head(keggres$less)
```

	p.geomean	stat.mean	p.val
hsa04110 Cell cycle	8.995727e-06	-4.378644	8.995727e-06
hsa03030 DNA replication	9.424076e-05	-3.951803	9.424076e-05
hsa03013 RNA transport	1.375901e-03	-3.028500	1.375901e-03
hsa03440 Homologous recombination	3.066756e-03	-2.852899	3.066756e-03
hsa04114 Oocyte meiosis	3.784520e-03	-2.698128	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	8.961413e-03	-2.405398	8.961413e-03

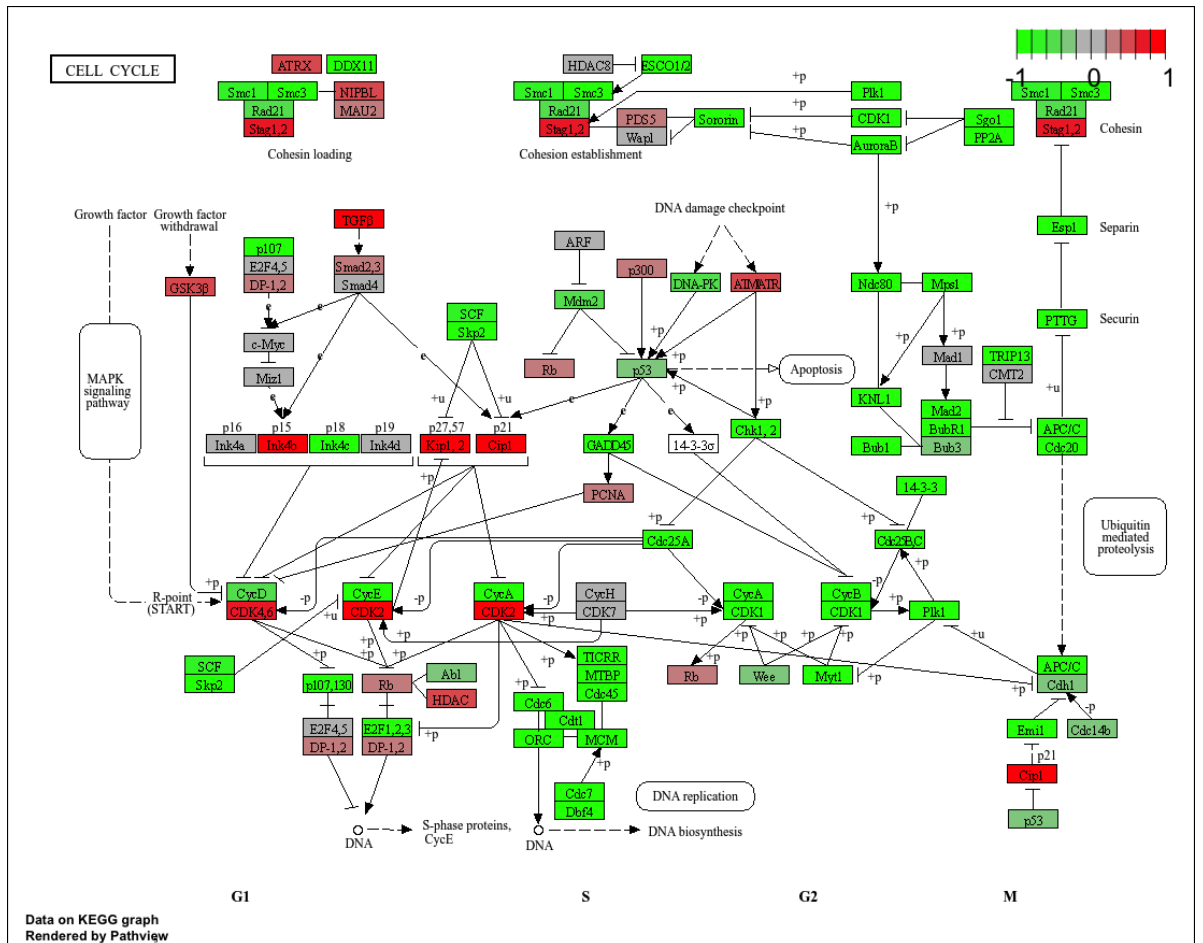
	q.val	set.size	exp1
hsa04110 Cell cycle	0.001448312	121	8.995727e-06
hsa03030 DNA replication	0.007586381	36	9.424076e-05
hsa03013 RNA transport	0.073840037	144	1.375901e-03
hsa03440 Homologous recombination	0.121861535	28	3.066756e-03
hsa04114 Oocyte meiosis	0.121861535	102	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	0.212222694	53	8.961413e-03

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/brianwong/Downloads/BIMM 143/class14

Info: Writing image file hsa04110.pathview.png



```
# A different PDF based output of the same data
```

```
pathview(gene.data=foldchanges, pathway.id="hsa04110", kegg.native=FALSE)
```

'select()' returned 1:1 mapping between keys and columns

Warning: reconcile groups sharing member nodes!

```
      [,1] [,2]  
[1,] "9"  "300"  
[2,] "9"  "306"
```

Info: Working in directory /Users/brianwong/Downloads/BIMM 143/class14

Info: Writing image file hsa04110.pathview.pdf

```
## Focus on top 5 upregulated pathways here for demo purposes only  
keggrespathways <- rownames(keggres$greater)[1:5]  
  
# Extract the 8 character long IDs part of each string  
keggresids = substr(keggrespathways, start=1, stop=8)  
keggresids
```

```
[1] "hsa04640" "hsa04630" "hsa00140" "hsa04142" "hsa04330"
```

```
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/brianwong/Downloads/BIMM 143/class14

Info: Writing image file hsa04640.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/brianwong/Downloads/BIMM 143/class14

Info: Writing image file hsa04630.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/brianwong/Downloads/BIMM 143/class14

Info: Writing image file hsa00140.pathview.png

'select()' returned 1:1 mapping between keys and columns

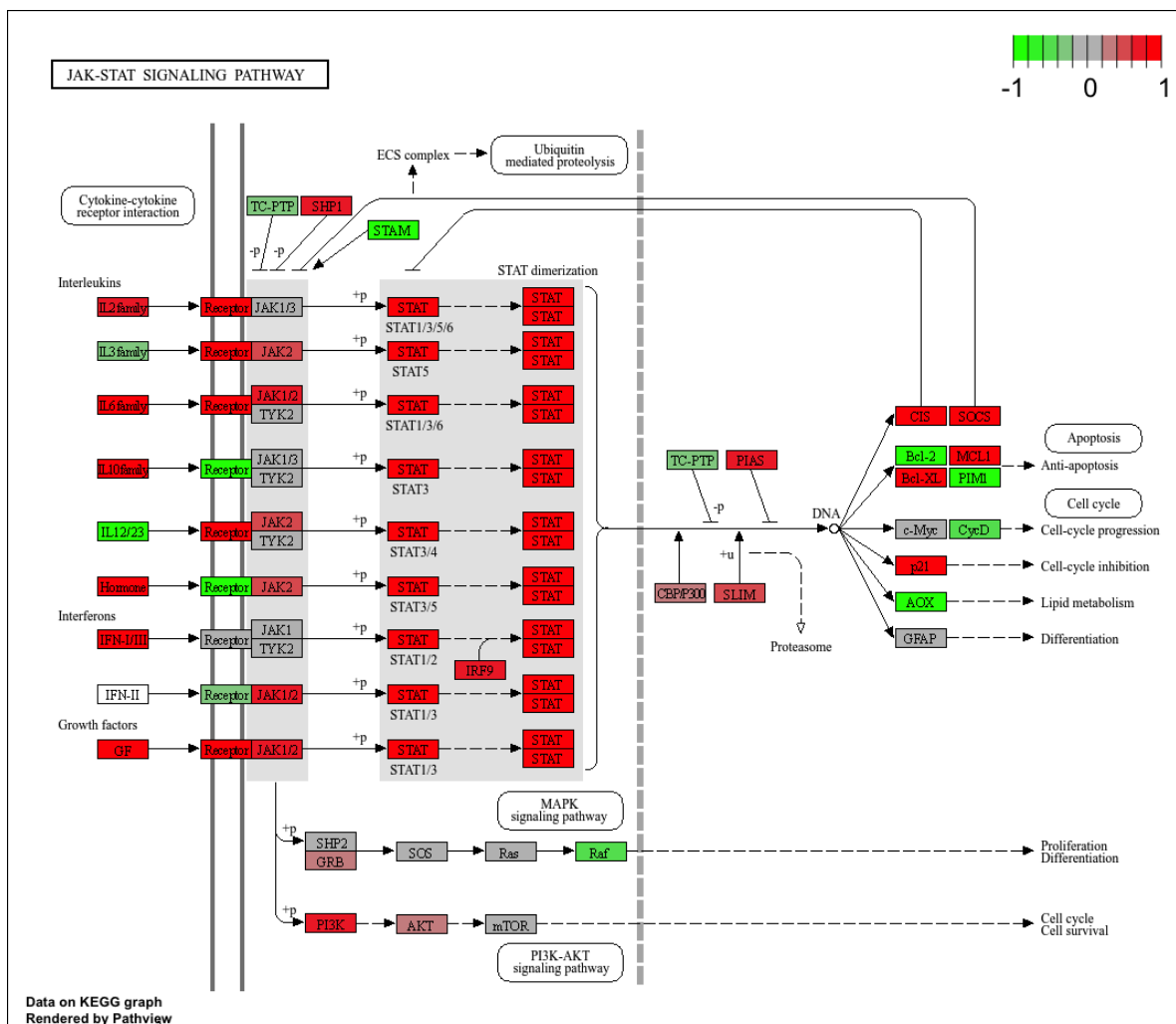
Info: Working in directory /Users/brianwong/Downloads/BIMM 143/class14

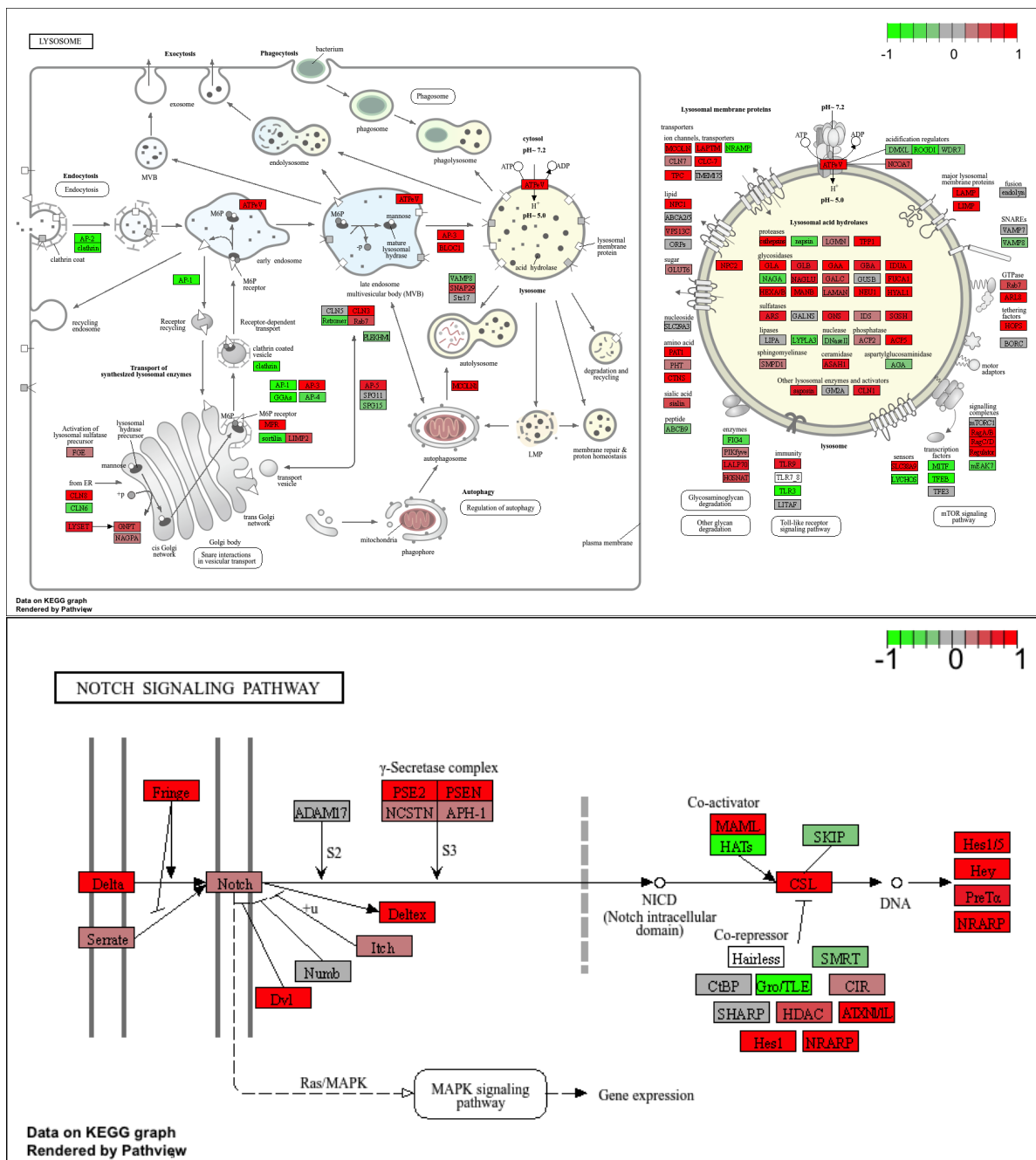
Info: Writing image file hsa04142.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/brianwong/Downloads/BIMM 143/class14

Info: Writing image file hsa04330.pathview.png





Q. Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

```
## Focus on top 5 down-regulated pathways
keggrespathways2 <- rownames(keggres$less)[1:5]
```

```
# Extract the 8 character long IDs part of each string
keggresids2 = substr(keggrespathways2, start=1, stop=8)
keggresids2
```

```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa03440" "hsa04114"
```

```
pathview(gene.data=foldchanges, pathway.id=keggresids2, species="hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/brianwong/Downloads/BIMM 143/class14

Info: Writing image file hsa04110.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/brianwong/Downloads/BIMM 143/class14

Info: Writing image file hsa03030.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/brianwong/Downloads/BIMM 143/class14

Info: Writing image file hsa03013.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/brianwong/Downloads/BIMM 143/class14

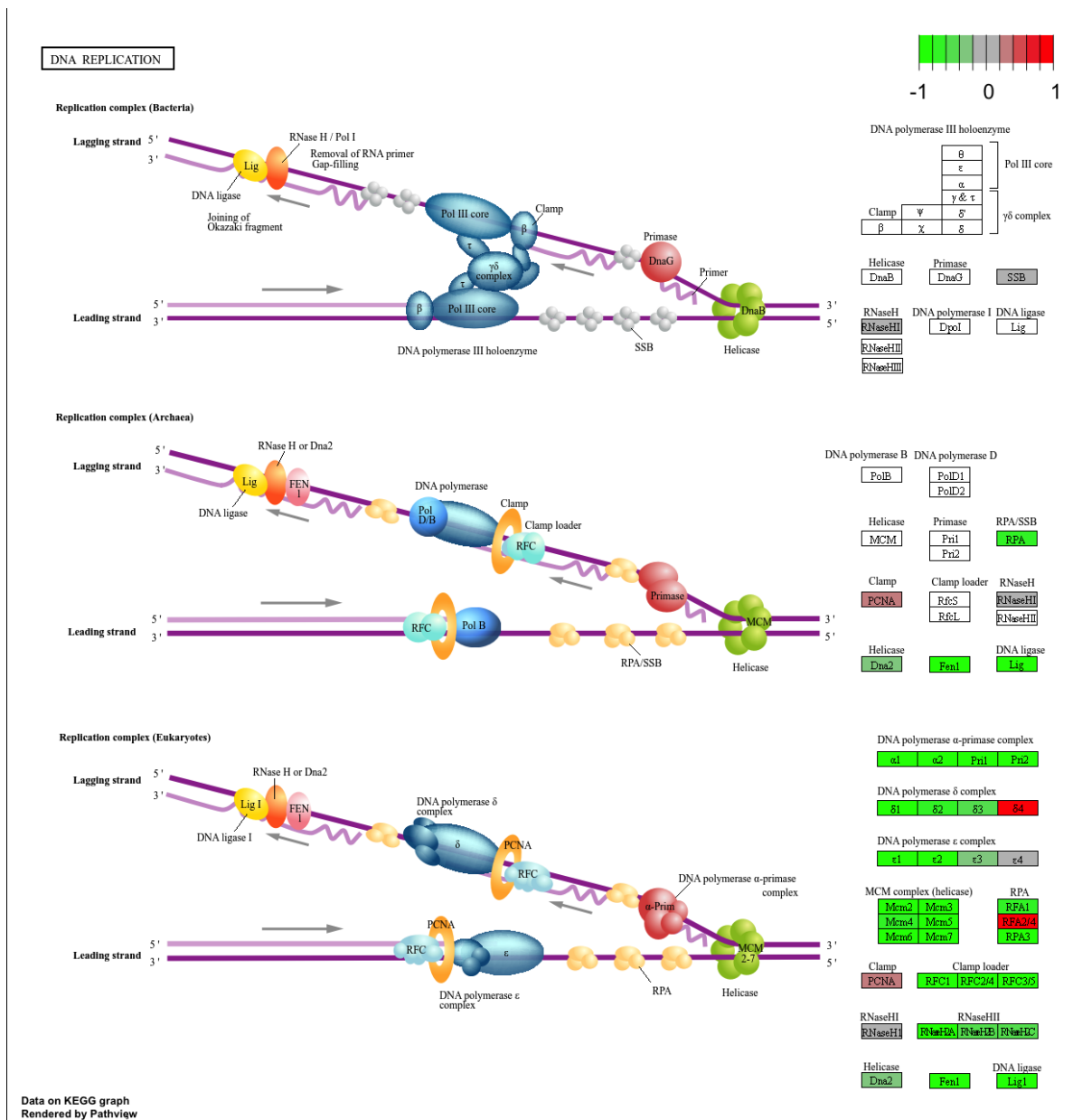
Info: Writing image file hsa03440.pathview.png

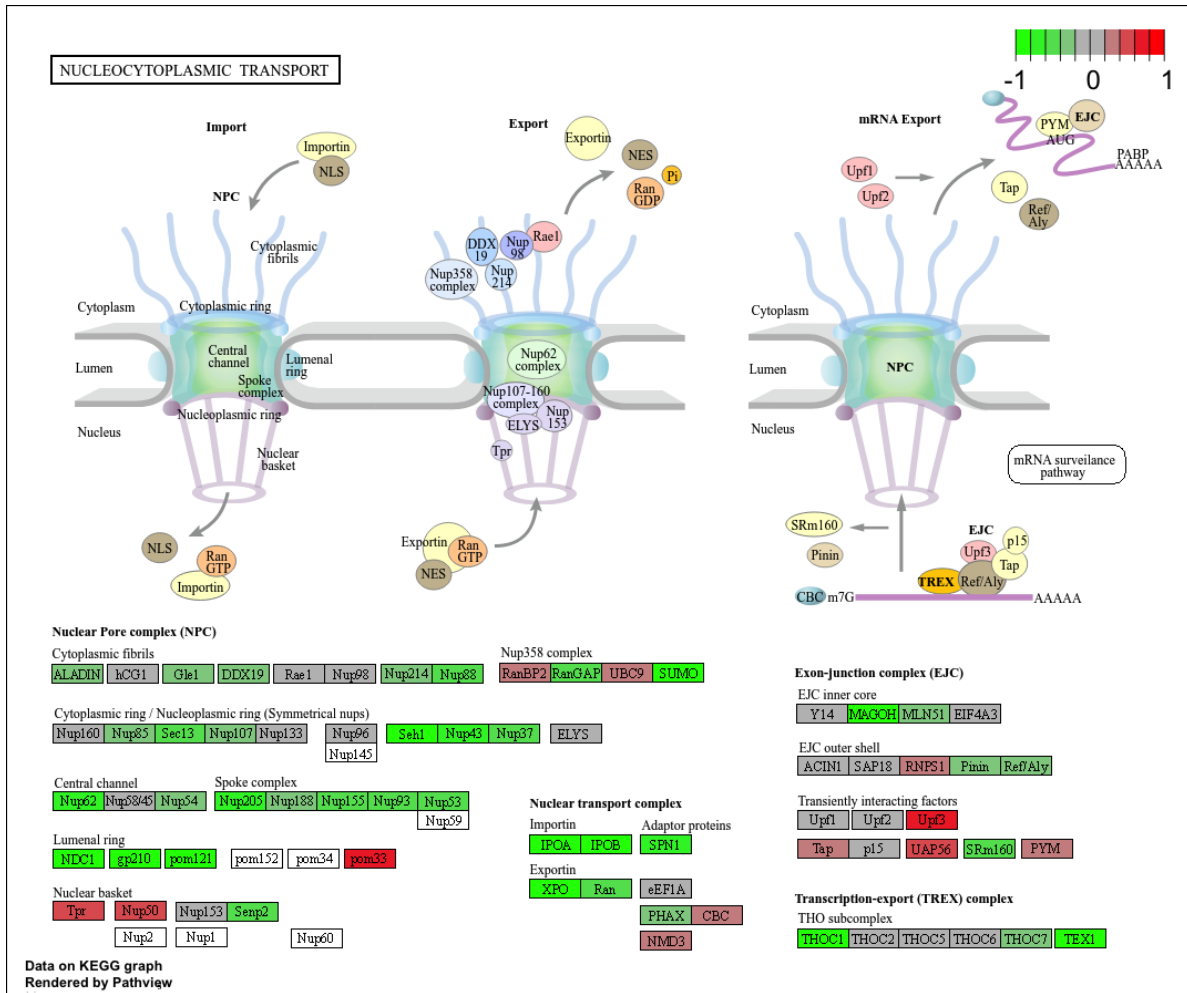
'select()' returned 1:1 mapping between keys and columns

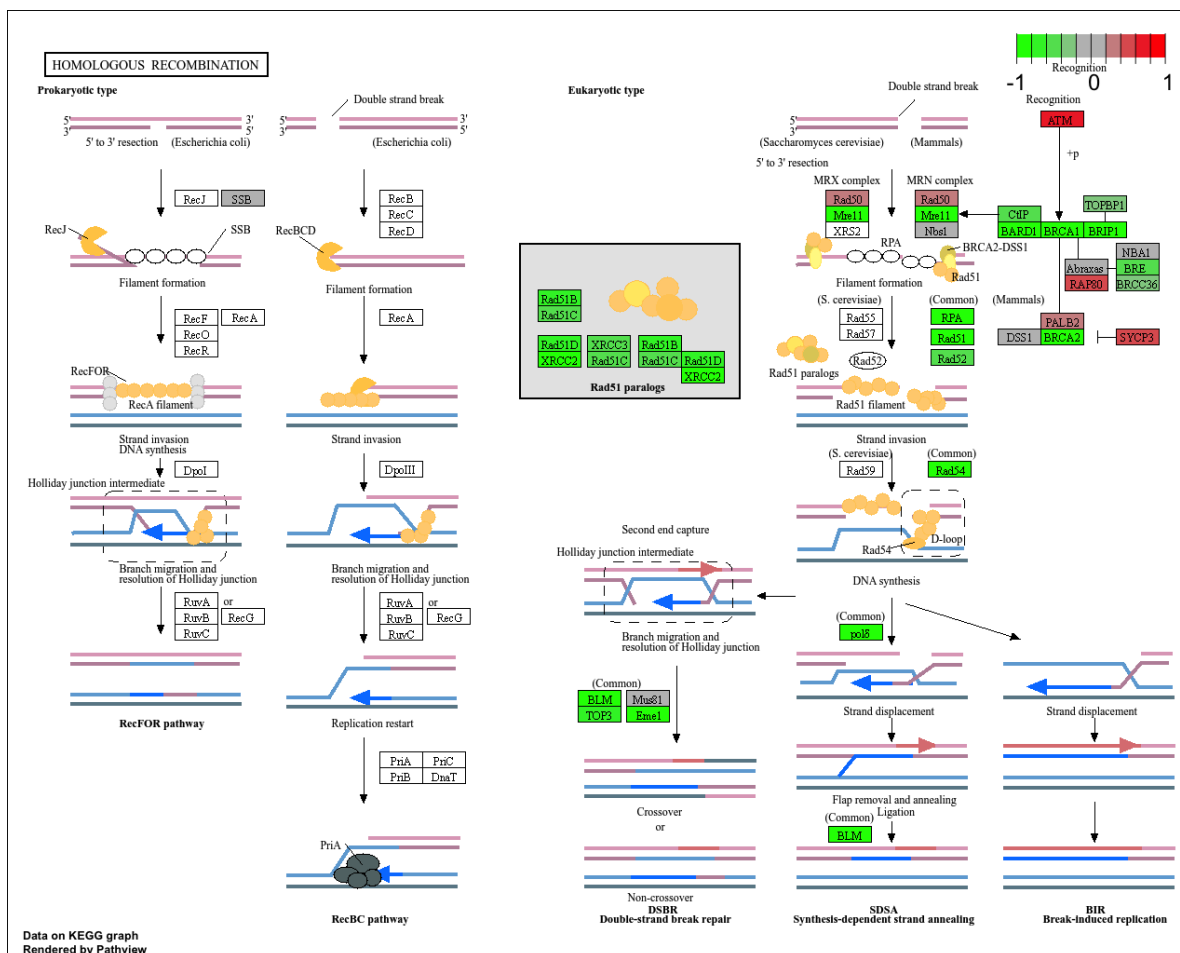
Info: Working in directory /Users/brianwong/Downloads/BIMM 143/class14

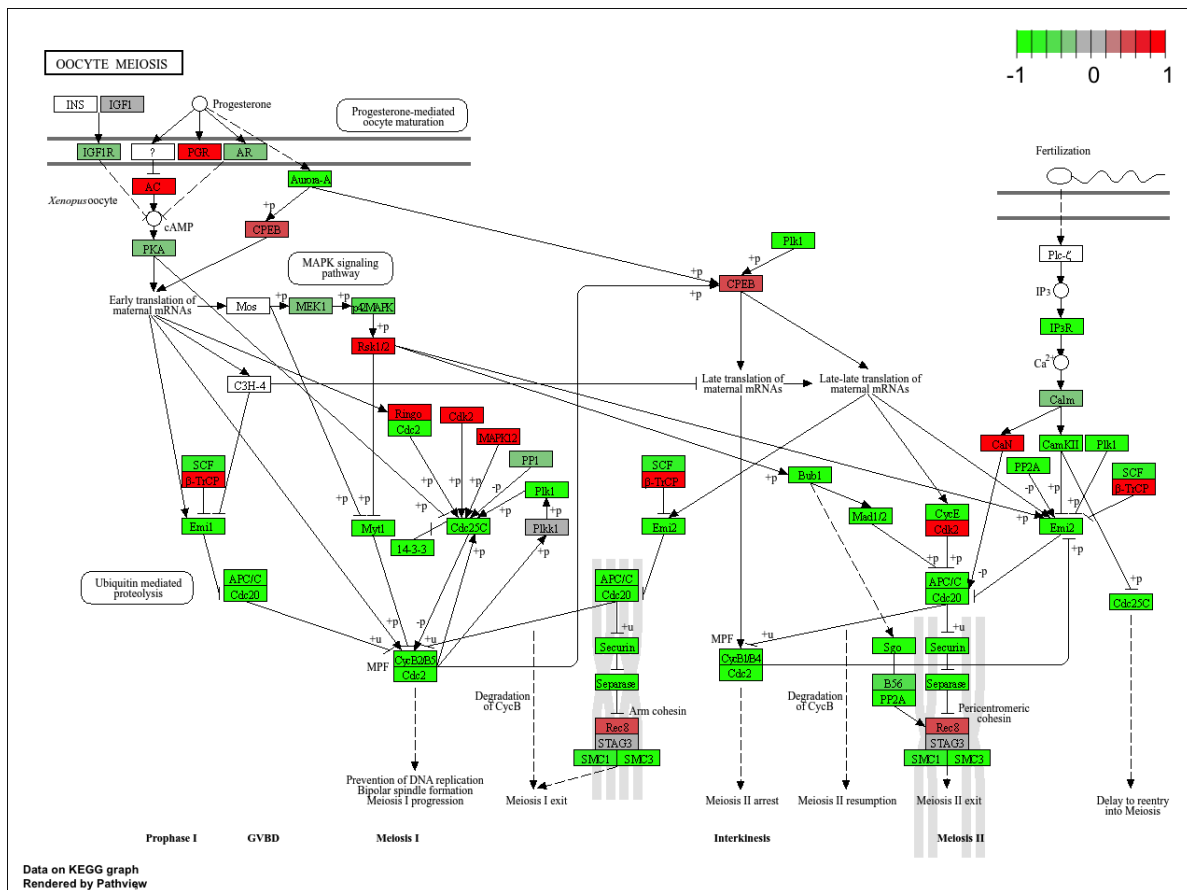
The diagram illustrates the cell cycle signaling pathways, showing the progression from G1 to S to G2 to M phases. Key components and interactions include:

- G1 Phase:** Growth factor withdrawal leads to GSK3 β activation, which inhibits c-Myc and Mdm2. c-Myc promotes p16, p15, p18, and p19, which inhibit Rb. Mdm2 inhibits p53. p53 promotes p21 and p27, which inhibit cyclins. The SCF complex (Skp2) promotes p21 and p27.
- S Phase:** DNA replication is initiated by the SCF complex (Skp2) and the E2F1/2/3 complex. The E2F1/2/3 complex is regulated by p107, p130, and Rb. The E2F1/2/3 complex promotes p107, p130, and Rb. The E2F1/2/3 complex is inhibited by p107, p130, and Rb. The E2F1/2/3 complex promotes p107, p130, and Rb. The E2F1/2/3 complex is inhibited by p107, p130, and Rb.
- G2 Phase:** DNA damage checkpoint is activated by DNA-PK and ATM/ATR, which phosphorylate p53. p53 promotes p21 and p27, which inhibit cyclins. The SCF complex (Skp2) promotes p21 and p27. The SCF complex (Skp2) promotes p21 and p27.
- M Phase:** The APC/C complex (Cdh1, Cdh2, Cdh3, Cdh4, Cdh5, Cdh6, Cdh7, Cdh8, Cdh9, Cdh10, Cdh11, Cdh12, Cdh13, Cdh14, Cdh15, Cdh16, Cdh17, Cdh18, Cdh19, Cdh20, Cdh21, Cdh22, Cdh23, Cdh24, Cdh25, Cdh26, Cdh27, Cdh28, Cdh29, Cdh30, Cdh31, Cdh32, Cdh33, Cdh34, Cdh35, Cdh36, Cdh37, Cdh38, Cdh39, Cdh40, Cdh41, Cdh42, Cdh43, Cdh44, Cdh45, Cdh46, Cdh47, Cdh48, Cdh49, Cdh50, Cdh51, Cdh52, Cdh53, Cdh54, Cdh55, Cdh56, Cdh57, Cdh58, Cdh59, Cdh60, Cdh61, Cdh62, Cdh63, Cdh64, Cdh65, Cdh66, Cdh67, Cdh68, Cdh69, Cdh70, Cdh71, Cdh72, Cdh73, Cdh74, Cdh75, Cdh76, Cdh77, Cdh78, Cdh79, Cdh80, Cdh81, Cdh82, Cdh83, Cdh84, Cdh85, Cdh86, Cdh87, Cdh88, Cdh89, Cdh90, Cdh91, Cdh92, Cdh93, Cdh94, Cdh95, Cdh96, Cdh97, Cdh98, Cdh99, Cdh100, Cdh101, Cdh102, Cdh103, Cdh104, Cdh105, Cdh106, Cdh107, Cdh108, Cdh109, Cdh110, Cdh111, Cdh112, Cdh113, Cdh114, Cdh115, Cdh116, Cdh117, Cdh118, Cdh119, Cdh120, Cdh121, Cdh122, Cdh123, Cdh124, Cdh125, Cdh126, Cdh127, Cdh128, Cdh129, Cdh130, Cdh131, Cdh132, Cdh133, Cdh134, Cdh135, Cdh136, Cdh137, Cdh138, Cdh139, Cdh140, Cdh141, Cdh142, Cdh143, Cdh144, Cdh145, Cdh146, Cdh147, Cdh148, Cdh149, Cdh150, Cdh151, Cdh152, Cdh153, Cdh154, Cdh155, Cdh156, Cdh157, Cdh158, Cdh159, Cdh160, Cdh161, Cdh162, Cdh163, Cdh164, Cdh165, Cdh166, Cdh167, Cdh168, Cdh169, Cdh170, Cdh171, Cdh172, Cdh173, Cdh174, Cdh175, Cdh176, Cdh177, Cdh178, Cdh179, Cdh180, Cdh181, Cdh182, Cdh183, Cdh184, Cdh185, Cdh186, Cdh187, Cdh188, Cdh189, Cdh190, Cdh191, Cdh192, Cdh193, Cdh194, Cdh195, Cdh196, Cdh197, Cdh198, Cdh199, Cdh200, Cdh201, Cdh202, Cdh203, Cdh204, Cdh205, Cdh206, Cdh207, Cdh208, Cdh209, Cdh210, Cdh211, Cdh212, Cdh213, Cdh214, Cdh215, Cdh216, Cdh217, Cdh218, Cdh219, Cdh220, Cdh221, Cdh222, Cdh223, Cdh224, Cdh225, Cdh226, Cdh227, Cdh228, Cdh229, Cdh230, Cdh231, Cdh232, Cdh233, Cdh234, Cdh235, Cdh236, Cdh237, Cdh238, Cdh239, Cdh240, Cdh241, Cdh242, Cdh243, Cdh244, Cdh245, Cdh246, Cdh247, Cdh248, Cdh249, Cdh250, Cdh251, Cdh252, Cdh253, Cdh254, Cdh255, Cdh256, Cdh257, Cdh258, Cdh259, Cdh260, Cdh261, Cdh262, Cdh263, Cdh264, Cdh265, Cdh266, Cdh267, Cdh268, Cdh269, Cdh270, Cdh271, Cdh272, Cdh273, Cdh274, Cdh275, Cdh276, Cdh277, Cdh278, Cdh279, Cdh280, Cdh281, Cdh282, Cdh283, Cdh284, Cdh285, Cdh286, Cdh287, Cdh288, Cdh289, Cdh290, Cdh291, Cdh292, Cdh293, Cdh294, Cdh295, Cdh296, Cdh297, Cdh298, Cdh299, Cdh300, Cdh301, Cdh302, Cdh303, Cdh304, Cdh305, Cdh306, Cdh307, Cdh308, Cdh309, Cdh310, Cdh311, Cdh312, Cdh313, Cdh314, Cdh315, Cdh316, Cdh317, Cdh318, Cdh319, Cdh320, Cdh321, Cdh322, Cdh323, Cdh324, Cdh325, Cdh326, Cdh327, Cdh328, Cdh329, Cdh330, Cdh331, Cdh332, Cdh333, Cdh334, Cdh335, Cdh336, Cdh337, Cdh338, Cdh339, Cdh340, Cdh341, Cdh342, Cdh343, Cdh344, Cdh345, Cdh346, Cdh347, Cdh348, Cdh349, Cdh350, Cdh351, Cdh352, Cdh353, Cdh354, Cdh355, Cdh356, Cdh357, Cdh358, Cdh359, Cdh360, Cdh361, Cdh362, Cdh363, Cdh364, Cdh365, Cdh366, Cdh367, Cdh368, Cdh369, Cdh370, Cdh371, Cdh372, Cdh373, Cdh374, Cdh375, Cdh376, Cdh377, Cdh378, Cdh379, Cdh380, Cdh381, Cdh382, Cdh383, Cdh384, Cdh385, Cdh386, Cdh387, Cdh388, Cdh389, Cdh390, Cdh391, Cdh392, Cdh393, Cdh394, Cdh395, Cdh396, Cdh397, Cdh398, Cdh399, Cdh400, Cdh401, Cdh402, Cdh403, Cdh404, Cdh405, Cdh406, Cdh407, Cdh408, Cdh409, Cdh410, Cdh411, Cdh412, Cdh413, Cdh414, Cdh415, Cdh416, Cdh417, Cdh418, Cdh419, Cdh420, Cdh421, Cdh422, Cdh423, Cdh424, Cdh425, Cdh426, Cdh427, Cdh428, Cdh429, Cdh430, Cdh431, Cdh432, Cdh433, Cdh434, Cdh435, Cdh436, Cdh437, Cdh438, Cdh439, Cdh440, Cdh441, Cdh442, Cdh443, Cdh444, Cdh445, Cdh446, Cdh447, Cdh448, Cdh449, Cdh450, Cdh451, Cdh452, Cdh453, Cdh454, Cdh455, Cdh456, Cdh457, Cdh458, Cdh459, Cdh460, Cdh461, Cdh462, Cdh463, Cdh464, Cdh465, Cdh466, Cdh467, Cdh468, Cdh469, Cdh470, Cdh471, Cdh472, Cdh473, Cdh474, Cdh475, Cdh476, Cdh477, Cdh478, Cdh479, Cdh480, Cdh481, Cdh482, Cdh483, Cdh484, Cdh485, Cdh486, Cdh487, Cdh488, Cdh489, Cdh490, Cdh491, Cdh492, Cdh493, Cdh494, Cdh495, Cdh496, Cdh497, Cdh498, Cdh499, Cdh500, Cdh501, Cdh502, Cdh503, Cdh504, Cdh505, Cdh506, Cdh507, Cdh508, Cdh509, Cdh510, Cdh511, Cdh512, Cdh513, Cdh514, Cdh515, Cdh516, Cdh517, Cdh518, Cdh519, Cdh520, Cdh521, Cdh522, Cdh523, Cdh524, Cdh525, Cdh526, Cdh527, Cdh528, Cdh529, Cdh530, Cdh531, Cdh532, Cdh533, Cdh534, Cdh535, Cdh536, Cdh537, Cdh538, Cdh539, Cdh540, Cdh541, Cdh542, Cdh543, Cdh544, Cdh545, Cdh546, Cdh547, Cdh548, Cdh549, Cdh550, Cdh551, Cdh552, Cdh553, Cdh554, Cdh555, Cdh556, Cdh557, Cdh558, Cdh559, Cdh560, Cdh561, Cdh562, Cdh563, Cdh564, Cdh565, Cdh566, Cdh567, Cdh568, Cdh569, Cdh570, Cdh571, Cdh572, Cdh573, Cdh574, Cdh575, Cdh576, Cdh577, Cdh578, Cdh579, Cdh580, Cdh581, Cdh582, Cdh583, Cdh584, Cdh585, Cdh586, Cdh587, Cdh588, Cdh589, Cdh590, Cdh591, Cdh592, Cdh593, Cdh594, Cdh595, Cdh596, Cdh597, Cdh598, Cdh599, Cdh600, Cdh601, Cdh602, Cdh603, Cdh604, Cdh605, Cdh606, Cdh607, Cdh608, Cdh609, Cdh610, Cdh611, Cdh612, Cdh613









Section 3. Gene Ontology (GO)

GO Analysis

Focus on the Biological Process “BP” Section of GO

```
data(go.sets.hs)
data(go.subs.hs)

# Focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets)

lapply(gobpres, head)
```

\$greater

	p.geomean	stat.mean	p.val
G0:0007156 homophilic cell adhesion	8.519724e-05	3.824205	8.519724e-05
G0:0002009 morphogenesis of an epithelium	1.396681e-04	3.653886	1.396681e-04
G0:0048729 tissue morphogenesis	1.432451e-04	3.643242	1.432451e-04
G0:0007610 behavior	1.925222e-04	3.565432	1.925222e-04
G0:0060562 epithelial tube morphogenesis	5.932837e-04	3.261376	5.932837e-04
G0:0035295 tube development	5.953254e-04	3.253665	5.953254e-04

	q.val	set.size	exp1
G0:0007156 homophilic cell adhesion	0.1951953	113	8.519724e-05
G0:0002009 morphogenesis of an epithelium	0.1951953	339	1.396681e-04
G0:0048729 tissue morphogenesis	0.1951953	424	1.432451e-04
G0:0007610 behavior	0.1967577	426	1.925222e-04
G0:0060562 epithelial tube morphogenesis	0.3565320	257	5.932837e-04
G0:0035295 tube development	0.3565320	391	5.953254e-04

\$less

	p.geomean	stat.mean	p.val
G0:0048285 organelle fission	1.536227e-15	-8.063910	1.536227e-15
G0:0000280 nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067 mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087 M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
G0:0007059 chromosome segregation	2.028624e-11	-6.878340	2.028624e-11
G0:0000236 mitotic prometaphase	1.729553e-10	-6.695966	1.729553e-10

	q.val	set.size	exp1
G0:0048285 organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280 nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067 mitosis	5.841698e-12	352	4.286961e-15
G0:0000087 M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14
G0:0007059 chromosome segregation	1.658603e-08	142	2.028624e-11
G0:0000236 mitotic prometaphase	1.178402e-07	84	1.729553e-10

\$stats

	stat.mean	exp1
G0:0007156 homophilic cell adhesion	3.824205	3.824205
G0:0002009 morphogenesis of an epithelium	3.653886	3.653886
G0:0048729 tissue morphogenesis	3.643242	3.643242
G0:0007610 behavior	3.565432	3.565432
G0:0060562 epithelial tube morphogenesis	3.261376	3.261376
G0:0035295 tube development	3.253665	3.253665

Section 4. Reactome Analysis

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]  
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

Q: What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The pathway that has the most significant “Entities p-value” is “Cell Cycle, Mitotic”. Not quite, the significant pathways on the previous KEGG results do not match completely to the significant pathways on Reactome. A factor that could cause the differences between the two methods are the different databases that these use to define their gene ontologies. Another factor is the different correction and statistical methods for measuring significance of genes and the different pathways.

Section 5: GO Online

Q: What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The pathway that has the most significant “Entities p-value” is “CENP-A containing chromatin assembly”. The significant pathways listed do not match the previous KEGG results and the factors that cause this difference could be explained like it was above with Reactome.